ARAB YOUTH SPACE HACKATHON 2026 · CHALLENGE 813 · FINAL

Blue Carbon Guardian

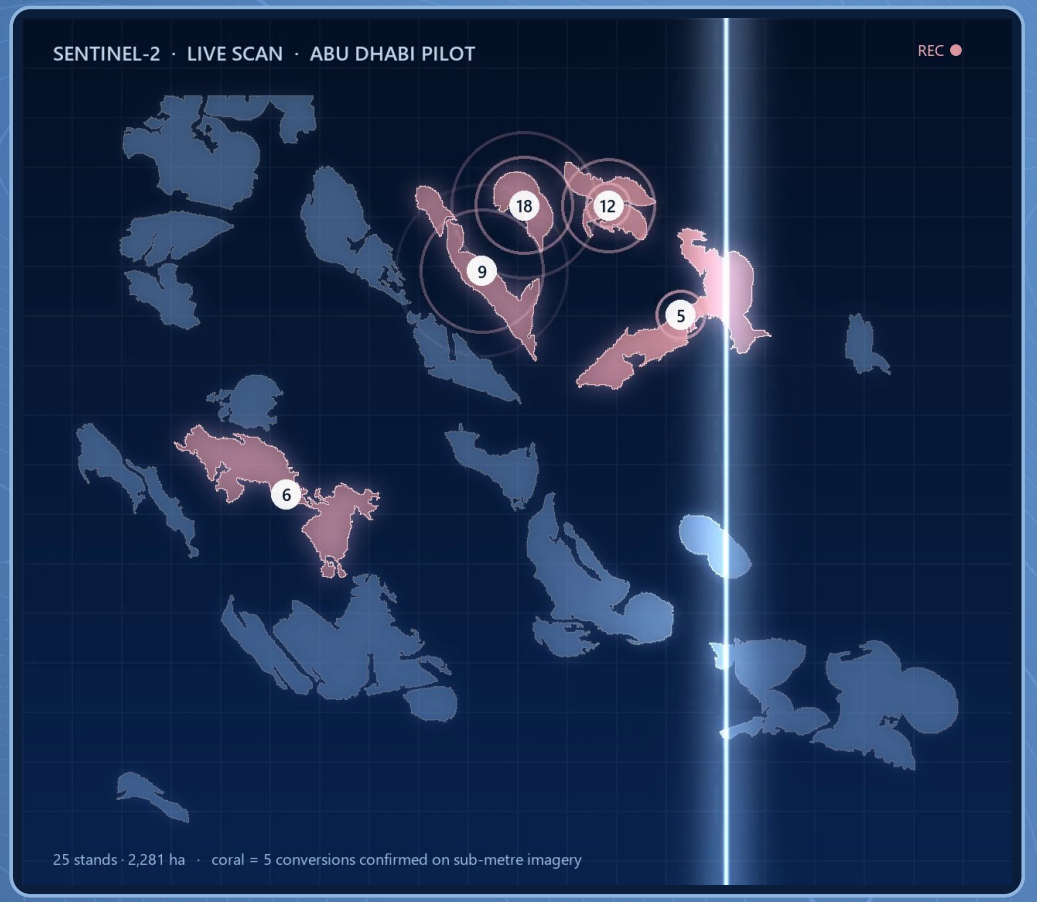
Tide-aware satellite monitoring for Gulf mangroves — from alert to evidence.

Team Blue Athar · الأثر الأزرق

Theme: Ecosystem Health, Biodiversity & Blue Carbon

91	5	73 %	≈ 1
stands monitored	conversions confirmed	partial loss caught (chance 13 %)	false alarm per stand-year

Nahla Nabil · Bahrain



Study area: Abu Dhabi, UAE

Built in Bahrain

THE TEAM

One engineer, every role — delivered in the open



Nahla Nabil

Team lead · AI engineer · IEEE researcher

Bahrain

EDUCATION

BSc Artificial Intelligence · Arab Open University, Bahrain · final semester · First-Class Honours standing

RELEVANT EXPERTISE

- 2 IEEE papers (ICCA 2025, ICAISET 2026): federated learning for industrial IoT; random-forest audit of GCC sustainability
- AI & LLM product engineering (Doo Bahrain; FlyRank backend AI)
- Machine learning, computer vision (deepfake detection), Python geospatial stack

COMMUNITY

NASA Space Apps Challenge local lead (Manama): organising and mentoring space-data teams

ROLES & RESPONSIBILITIES IN THIS PROJECT

Project lead & owner

problem framing, user needs, scope, every claim

EO data engineer

778 Sentinel-2 dates, EnMAP reader, sub-metre checks

Modelling & statistics

tide-aware model, stand + cell alert, calibration

Validation & integrity

real events, simulations, unseen stands

Product & delivery

dashboard, website, site reports, notebook

Business & impact

customers, pricing hypotheses, carbon ranges

Incubation gap we will close: a mangrove ecologist mentor · a restoration or agency partner · a GIS analyst

THE PROBLEM

Restoration is scaling faster than monitoring

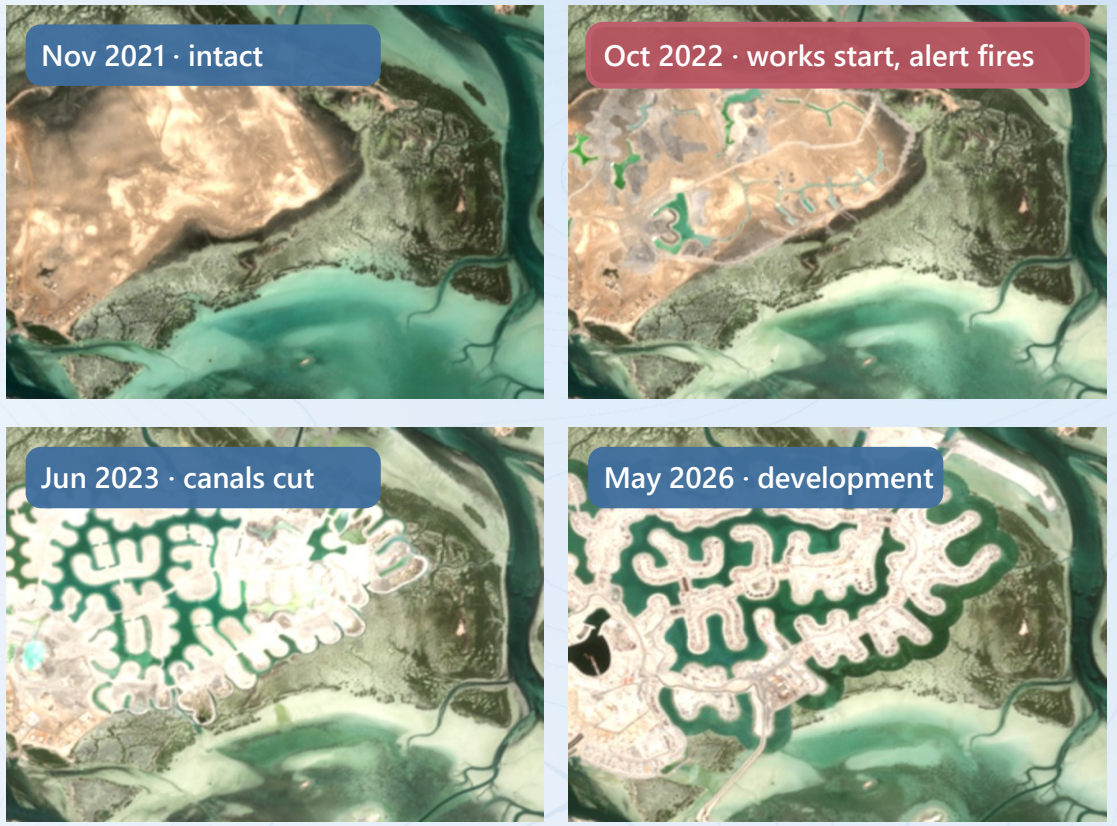
100 M — **UAE mangrove pledge by 2030**
COP26, 2021 · Abu Dhabi Mangrove Initiative

100 M+ — **Saudi mangrove pledge by 2030**
Saudi Green Initiative

Feb 2026 — **first GCC mangrove monitoring guide**
EAD + IUCN Mangrove Specialist Group

TODAY field surveys and one-off studies, no standing alert.
WE SOLVE a calibrated alert every ~5 days that says where to inspect and what carbon is at stake.

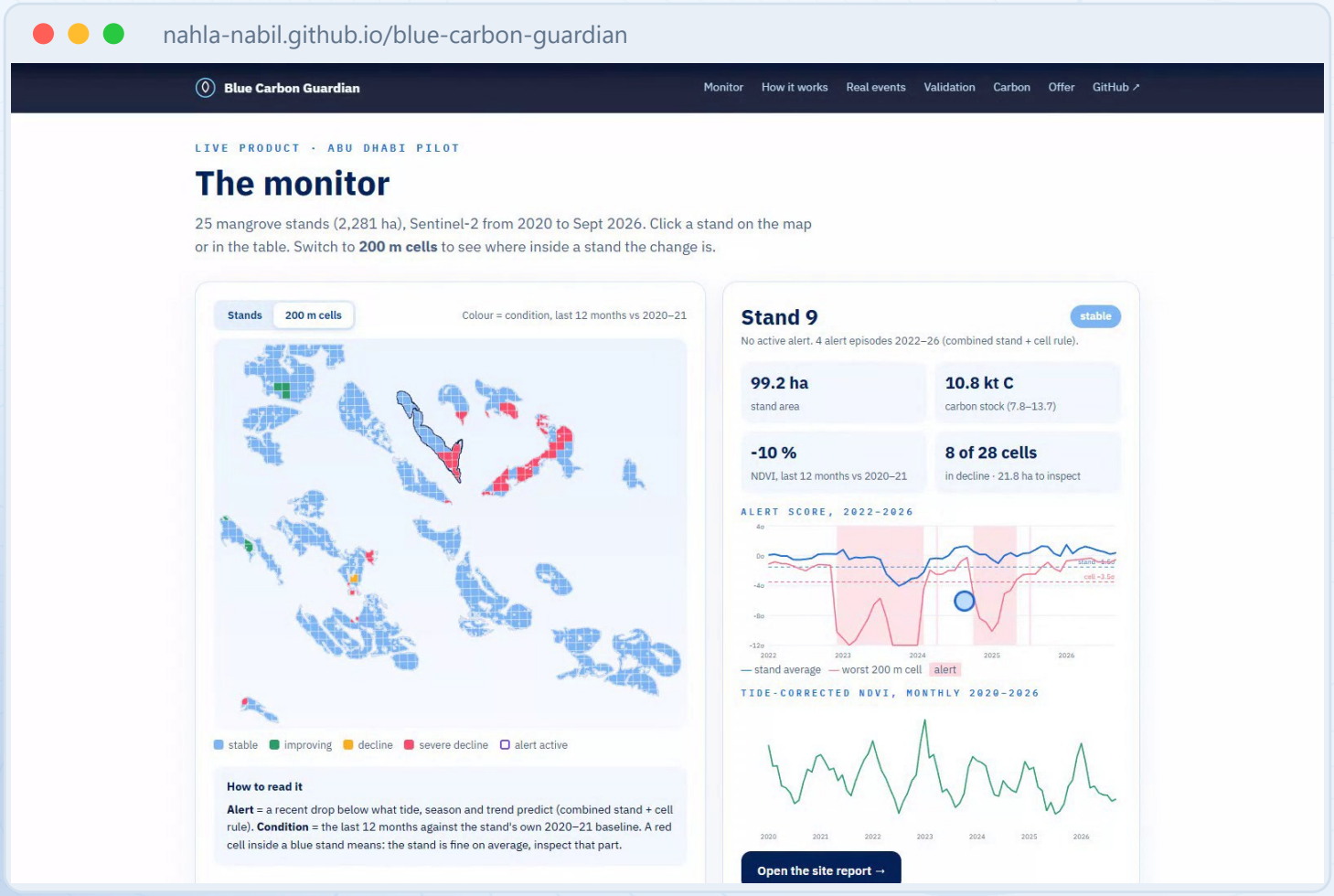
THIS IS HAPPENING NOW · STAND 5, ABU DHABI



Sentinel-2 true colour (contains modified Copernicus Sentinel data). Our alert began on 25 Oct 2022, during the works.

EXAMPLE OUTPUT · THE PRODUCT

From signal to decision, in one screen



One alert
stand OR any 200 m cell, calibrated

Where to look
a 200 m cell map for the inspector

Condition
last 12 months vs own 2020–21 baseline

Carbon at stake
stock with a range, from field data

Site report
one page per stand, every number traced

● LIVE DEMO · open stand 9 on the website

BUSINESS USE CASE & END USERS

Who uses it, and for which decision

Environment agencies

EAD, MOCCAE, Saudi NCVC

THEIR DECISION

Which stands do we inspect this month?

Today: periodic field surveys

monthly alert + 200 m cell map: inspect only flagged cells

Planting & carbon developers

ADNOC, Red Sea Global, carbon projects

THEIR DECISION

Is the carbon we report still there?

Today: plots and drone campaigns

condition + carbon stock with a range, a site report

Developers & regulators

EIA consultants, permit offices

THEIR DECISION

Did works stay outside the mangrove line?

Today: comparing two images by eye

an alert when works enter mangrove cells, dated evidence

New scene — every ~5 days → **Alert** — ~1 false alarm / stand-yr → **Cell map** — where to look → **Inspection** — only flagged cells → **Site report** — carbon at stake

Five independent data sources, six years, 91 stands



Pilot: 25 stands · 2,281 ha (WorldCover 2021 patches ≥ 3 ha). +66 unseen stands in Abu Dhabi N/W and Saudi Tarut Bay test transfer.

Sentinel-2 L2A
ESA / Copernicus · 10–20 m time series
778 dates

EnMAP hyperspectral
DLR · 224 bands · independent check
2 scenes

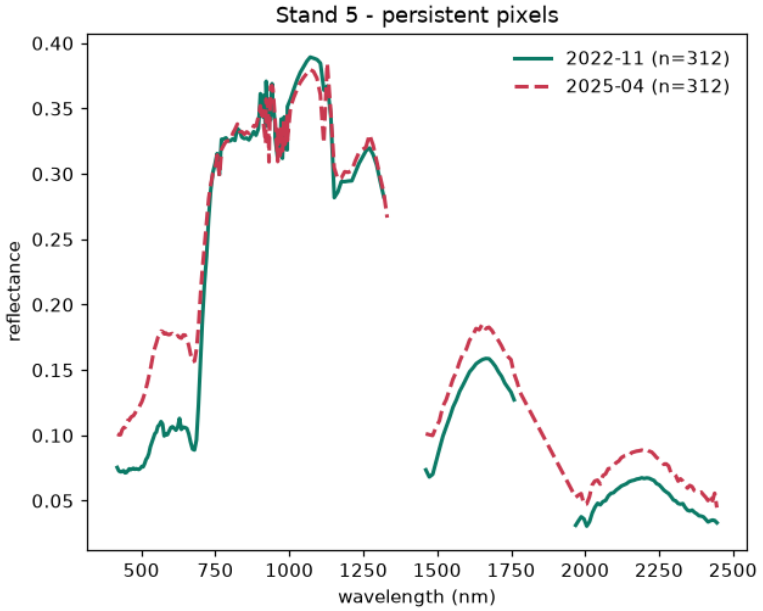
ESA WorldCover 2021
10 m land cover · stand outlines
CC BY 4.0

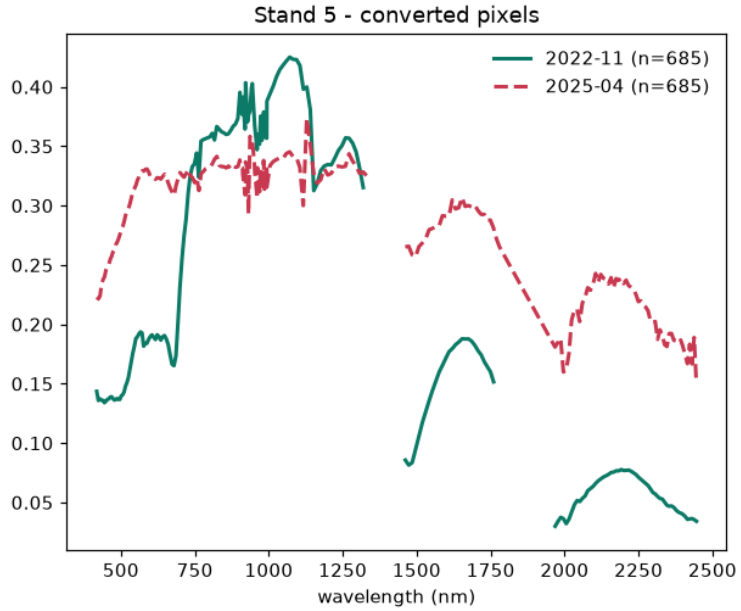
Field carbon plots
Schile et al. 2016 · carbon density
24 plots

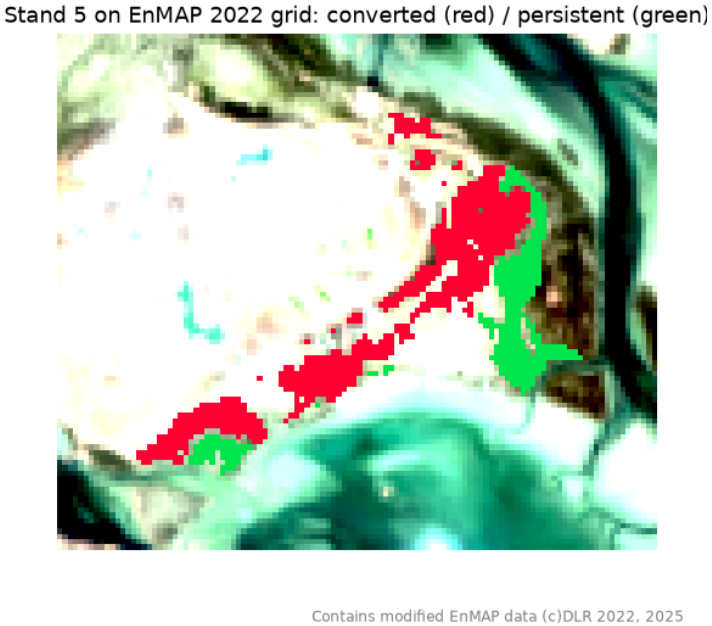
Sub-metre captures
WorldView-2/3, Legion-1 · confirm events
0.3–0.5 m

Quality control: cloud masks · interior pixels only · tide model · second-source checks

A second sensor shows what the mangrove became







r = 0.84–0.99

agreement with our Sentinel-2 indices on the same dates (25 and 14 stands)

62 vs 26 ha

stand-5 loss: EnMAP pixel count vs our conservative Sentinel-2 flag

Vegetation → mineral

converted pixels take the spectrum of sand and fill: reclamation, not dieback

Honest limit: as a classifier, the full spectrum is no better than Sentinel-2-like bands (F1 0.74 vs 0.72). Contains modified EnMAP data © DLR 2022, 2025.

SPACE DATA · READY FOR SATELLITE 813

Sentinel-2 says when. Satellite 813 says what.

Sensor	Bands	Range	Pixel	Revisit	Role in the product
Sentinel-2	13	443–2190 nm	10–20 m	~5 days	WHEN: the alert, every pass
EnMAP (used)	224	420–2450 nm	30 m	~27 days	WHAT: tested on stand 5
Satellite 813 (next)	**~205**	**~400–1700 nm**	**~20 m**	**on demand**	**WHAT, sharper: tasked on flagged cells**

1 · Sentinel-2 alert

stand or 200 m cell below its threshold

→

2 · Task Satellite 813

only over the flagged cells, on demand

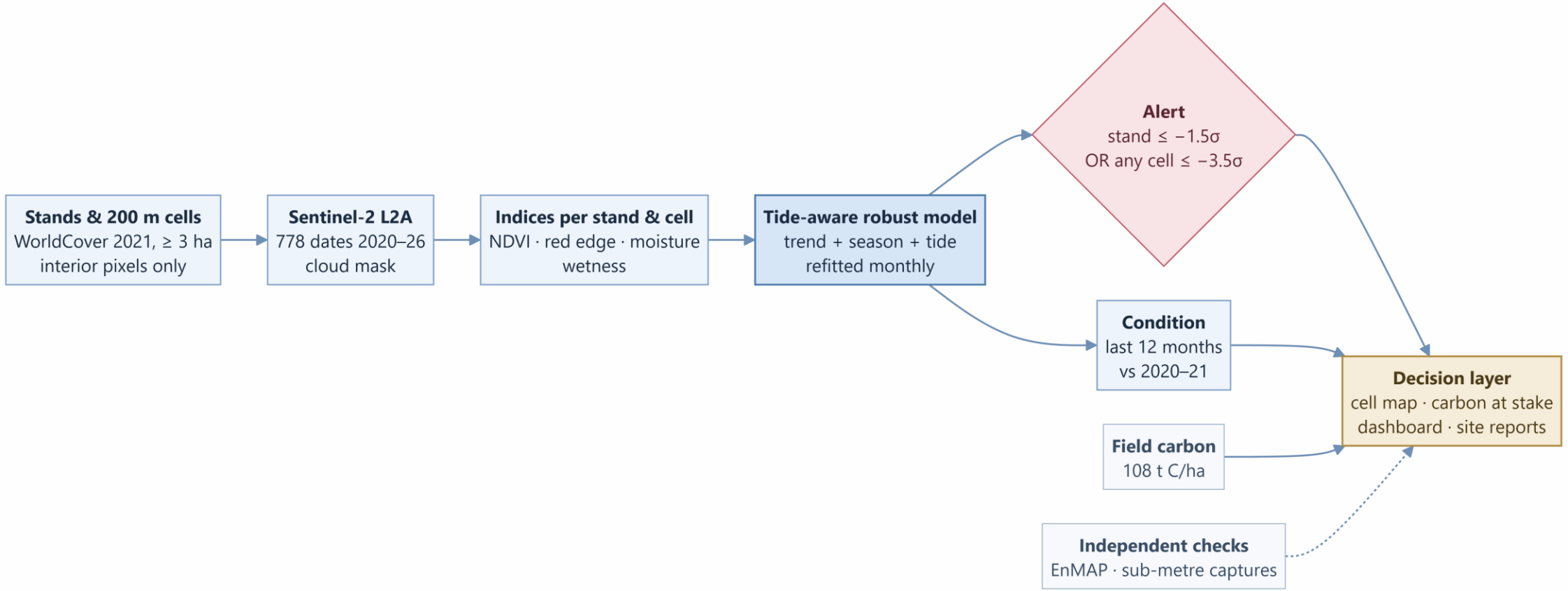
→

3 · Spectral diagnosis

vegetation, or sand and fill: an evidence pack

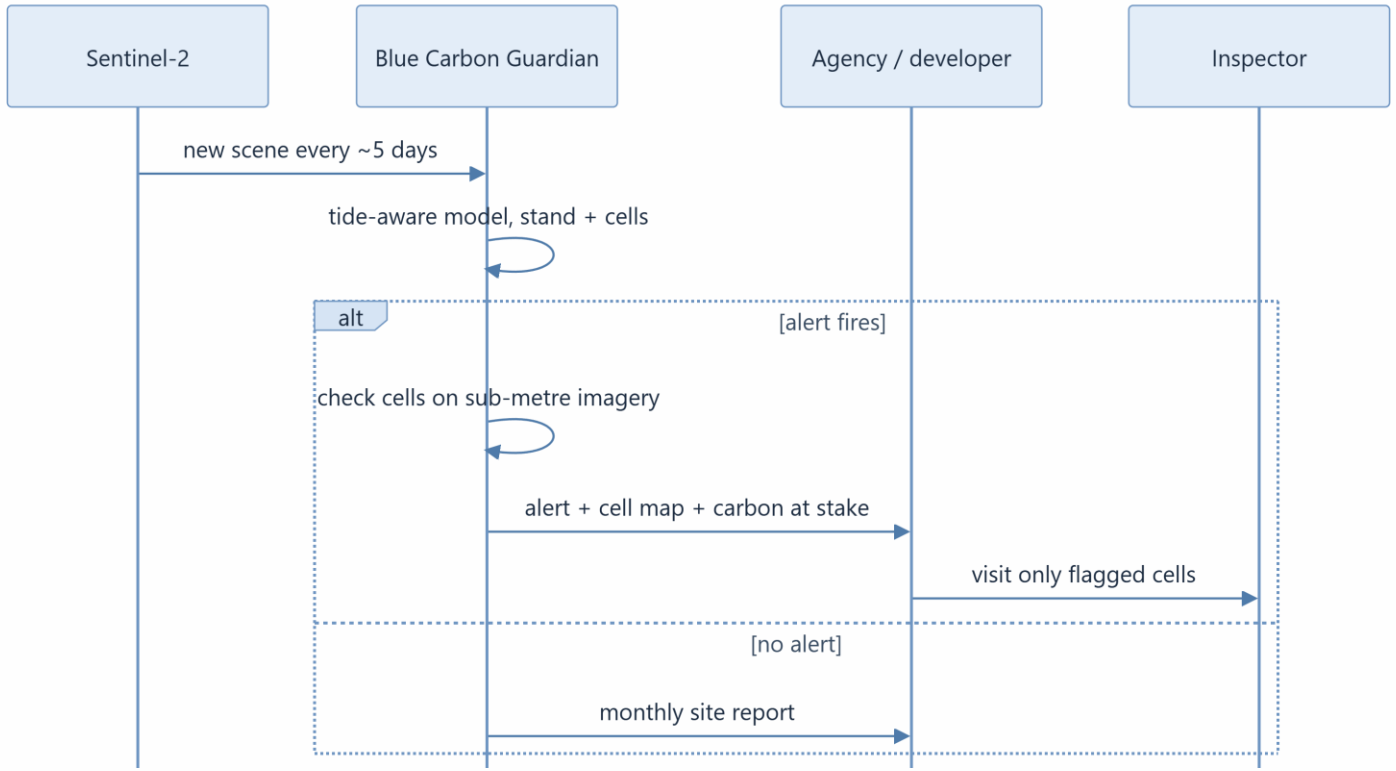
Every index the monitor uses (NDVI, red edge, SWIR moisture at 1610 nm, wetness) sits inside 813's ~400–1700 nm range, at 20 m instead of EnMAP's 30 m. **Planned, not yet tested:** 813 scenes were not available in the proof of concept.

From satellite pixels to a decision



Read left to right: Core model · Alert rule · Output to the user · Inputs and checks

One alert, one message, one targeted visit



Every ~5 days

a new Sentinel-2 scene is read in place and run through the tide-aware model, per stand and per 200 m cell

If an alert fires

we check the flagged cells on sub-metre imagery, then send the alert by email or JSON API with the cell map and the carbon at stake

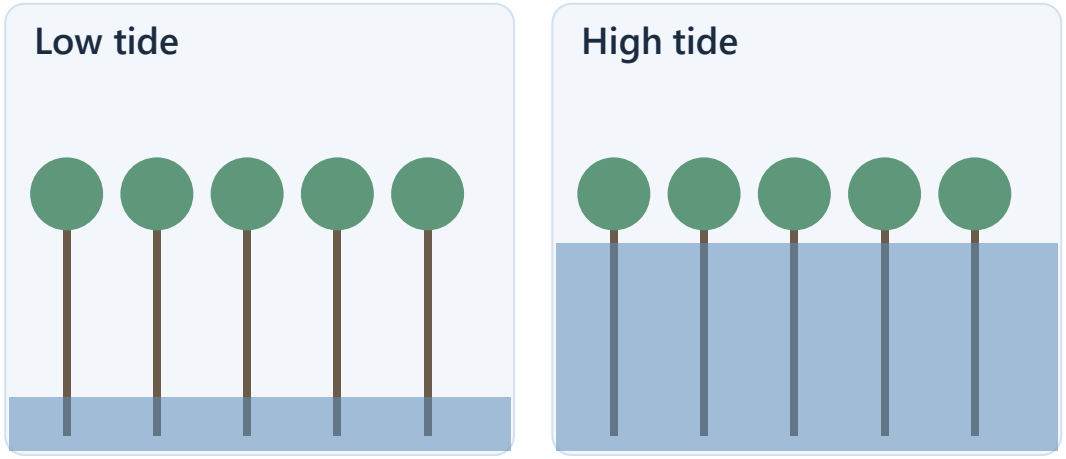
If nothing changes

the user still gets a monthly site report: condition, trend and carbon stock for every stand

TECHNICAL INNOVATION 1

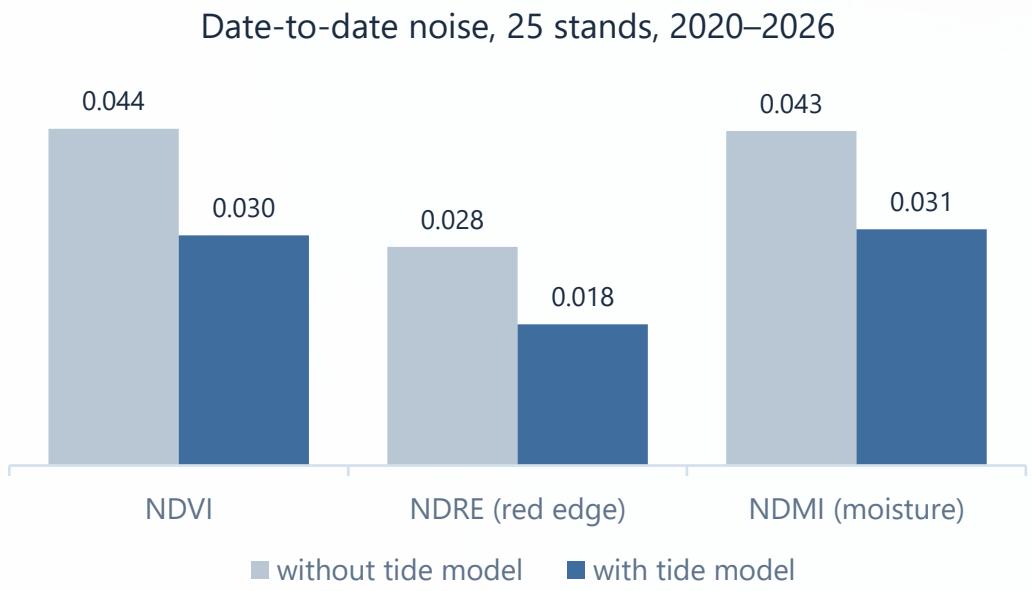
The tide looks like tree loss — until you correct for it

WHY NAIVE MONITORS FAIL ON TIDAL FLATS



NDVI **0.30** NDVI **0.18**

Same trees, same season. Measured on our stands: noise vs wetness r ≈ −0.9.



−29 to −35 %

less noise on every index: we model trend, season and the tide before flagging anything

1.61 → 0.96

false alarms per stand-year: textbook NDVI-only vs our tide-aware monitor, same alert rule, same 25 stands (40 % fewer)

A stand average hides partial loss. 200 m cells do not.



Real stand-5 data, 2022 → 2026: cells turn red while the stand average barely moves.

Partial loss caught within 60 days (simulated)

73 % stand + cells

25 % stand alone

13 % by chance

8 of 28

cells of stand 9 in severe decline while its average reads “stable”

3 of 5

real conversions found only by the cells (stands 12, 18, 6)

ONE ALERT RULE

stand ≤ −1.5σ OR any cell ≤ −3.5σ

calibrated together: 0.96 false-alarm episodes per stand-year

How we differ from what exists today

	Field surveys	One-off satellite maps	Free loss alerts (Global Mangrove Watch)	Blue Carbon Guardian
How often	campaigns	yearly or less	monthly	**every ~5 days**
Corrects for the tide	n/a	not usually	not documented	**✓ tide model**
Says where inside a stand	✓	—	—	**✓ 200 m cells**
Published false-alarm rate	n/a	n/a	accuracy only	**✓ ≈ 1 per stand-year**
Carbon at stake per alert	—	—	—	**✓ field-based**
Gulf coverage	✓	✓	not confirmed	**✓ Abu Dhabi + Tarut Bay**

Complementary, not competing: Global Mangrove Watch flags loss worldwide; we add the tide model, the 200 m cells, a measured false-alarm rate and the carbon at stake — built and tested for the Gulf.

Five real conversions, confirmed on sub-metre imagery



Stand 5 reclamation with finger canals



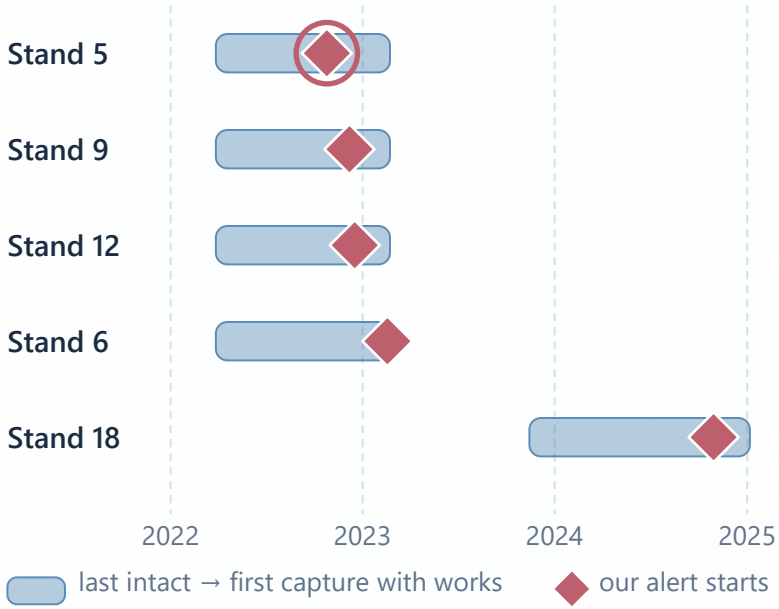
Stand 9 canal development, south-east



Stand 12 found only by the cells

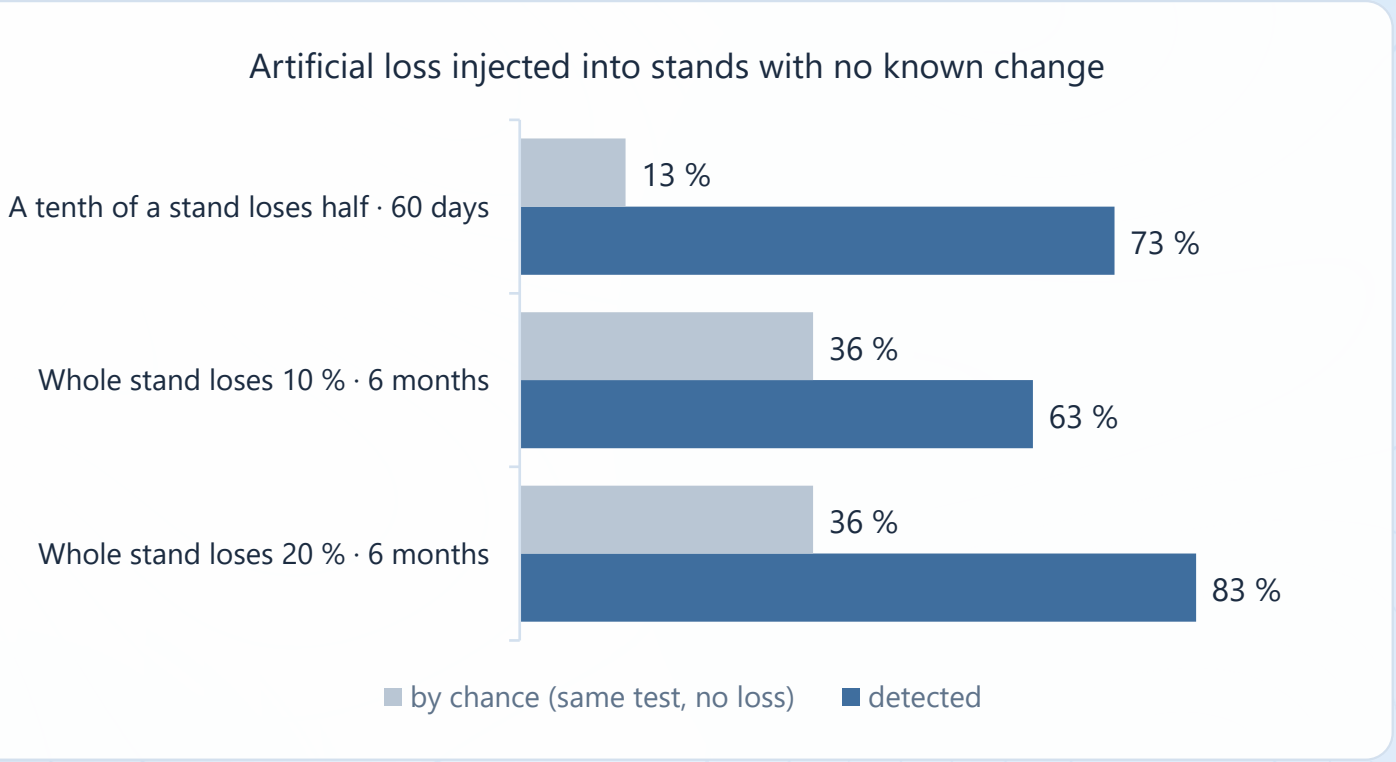


Stand 18 found only by the cells, 2025



Every alert began inside the window. Detection as works happen, not advance warning.

Every rate next to its chance rate



CALIBRATED · 25 STANDS

0.96

false-alarm episodes per stand-year

UNSEEN · 65 STANDS

0.79

per stand-year, 305 stand-years, incl. Saudi Tarut Bay

ENMAP · SAME DATES

r = 0.84–0.99

vs our Sentinel-2 indices; stand-5 loss 62 ha vs our 26 ha

A detection counts only if a new alert starts after the loss begins. ~160–240 trials per row.

What we do not claim: advance warning, field verification, species discrimination, or that hyperspectral classifies better (F1 0.74 vs 0.72). We withdrew our own first test (58 % / 90 %): it had no chance baseline.

Carbon at stake, and who pays to protect it

108 t C/ha

field density near the stands (90 % CI 79–138); 82 % of it is in the soil

247 kt C

pilot stock, 2,281 ha (range 180–316) ≈ 907 kt CO_2e

2.8–6.7 kt C

stock on the 26–62 ha lost at stand 5: what one alert puts on the table

SERVICE TIERS (PRICES ARE PILOT HYPOTHESES)

Monitor

$12k

per site / yr

alert, cell map, condition, carbon, monthly reports

Evidence

$25k

per site / yr

+ sub-metre check of every alert, audit pack

Development watch

$8k

per project / yr

for coastal developers and EIA consultants

Cost to serve ≈ $3.3k / site / yr: compute < $5 (measured), sub-metre checks $500–750, analyst $2k (assumption).

COMPETITION

Global Mangrove Watch sends free monthly loss alerts (Gulf coverage not confirmed). We are complementary: tide-aware, calibrated, stand + cell, carbon and reports.

THE HONEST MARKET

Gulf + Red Sea subscriptions ≈ $0.2–0.4 M / yr. The real prize: a **national MRV contract** or a monitoring layer on **gIQ**. SDG 13 · 14 · 15.

Reproducible today, built to scale tomorrow

BONUS · DOCUMENTATION & REPRODUCIBILITY

Public repository, MIT licence
code, notebook, figures, provenance and licences

Re-runs end to end in ~24 s
outputs committed · pinned requirements · example in/out

Clean-machine test passed
fresh environment, same numbers, 0 errors

Every number traceable
each figure comes from a script and a written summary

Live website + site reports
anyone can open the monitor and check a stand

BONUS · SCALABILITY BEYOND THE POC

Proven now
91 stands · 2 countries · 2 Sentinel-2 tiles · 65 unseen stands at 0.79 false alarms / stand-yr · compute < $5 per site per year

Incubation
Satellite 813 in the spectral step · deployment on gIQ · a pilot customer + site records · more EnMAP dates

Scale
Gulf ≈ 165 km^2 + Red Sea ≈ 200 km^2 of mangrove · sub-metre data for small stands (Bahrain) · national MRV, licensing

Areas: Frontiers in Marine Science 2025. Compute measured; prices assumed.

TEAM BLUE ATHAR · الأثر الأزرق

Every lost mangrove leaves a trace. We find it from space.

Overall value: a working product · five real events confirmed · every number reproducible

Tide-aware	5 of 5	0.79	Open
29–35 % less noise	conversions alerted during the works, 3 only by cells	false alarms / stand-yr on unseen stands	public code, re-runs in ~24 s

Our ask: incubation to run a pilot with a Gulf agency or planting programme, mentor access on hyperspectral reading of mangroves, and Satellite 813 scenes when available.

nahla-nabil.github.io/blue-carbon-guardian · github.com/Nahla-Nabil/blue-carbon-guardian



Thank you



Thank you

شكراً للإمارات

To the United Arab Emirates — for leading the region in protecting its mangroves, and for opening space to Arab youth.

With thanks to: UAE Space Agency · National Space Academy · Space42 · the mentors and the judges



From Bahrain, with gratitude — Nahla Nabil · Team Blue Athar

BACKUP · FOR QUESTIONS

The textbook approach against ours, same alert rule

Configuration	Indices	Tide model	False alarms per stand-year	Real conversions caught
A · textbook	NDVI	no	1.61	5 of 5
B	NDVI	yes	1.04	5 of 5
C	NDVI + red edge + SWIR moisture	no	1.36	5 of 5
D · ours	NDVI + red edge + SWIR moisture	yes	0.96	5 of 5

40 % fewer false alarms

20 stands with no known change, 93.9 stand-years: about 16 fewer unnecessary inspections a year for 25 stands.

The tide model is most of the gain

NDVI alone 1.61 → with tide terms 1.04; the extra indices bring it to 0.96.

Honest limit

Thresholds calibrated for our monitor; all four catch the five conversions. The difference is the false-alarm cost, not detection.

Where it breaks — and what we do not claim

Detection, not early warning

Alerts fire as works happen. We never claim to see stress before it is visible.

10 m pixels

Patches under ~0.6 ha and young plantings are missed, Bahrain's small stands included.

Imagery, not site visits

Conversions confirmed on dated sub-metre captures, not yet on permits or field records.

Simulated losses

Canopy-cover loss, not leaf stress; every rate shown next to its chance rate.

Hyperspectral

EnMAP confirms our indices but was not a better classifier (F1 0.74 vs 0.72).

Carbon

Area × field density from four sites (2013–14): a stock with a range, never a sequestration rate.

Rule we kept: no detection rate without its chance rate; no claim of advance warning, field verification or species discrimination.

The 2-minute product demo

